Machine Learning Powered Mosquito Population Control System for Sewage Networks

1. Alvin Christopher Chow, Canisius College, Jakarta, Indonesia (2415514alvin@kanisius.sch.id)
2. Benedict Halim, Canisius College, Jakarta, Indonesia (2415517benedict@kanisius.sch.id)
3. Denisson Setiawan Tanuraharja, Canisius College, Jakarta, Indonesia (2415451denisson@kanisius.sch.id)
4. Jonathan Soetaputra, Canisius College, Jakarta, Indonesia (2415595jonathan@kanisius.sch.id)
5. Kevan Baswaraputra Yusgiantoro, Canisius College, Jakarta, Indonesia (2415662kevan@kanisius.sch.id)
6. Rafael Lewis Tirtaatmadja, Canisius College, Jakarta, Indonesia (2415502rafael@kanisius.sch.id)

Abstract

Mosquito-borne diseases, particularly dengue and malaria, represent a severe and escalating public health crisis in Indonesia, contributing significantly to morbidity and mortality rates. Conventional vector control strategies are increasingly compromised by widespread insecticide resistance and operational limitations, necessitating a paradigm shift towards innovative, sustainable solutions. This paper presents the design and architecture of an autonomous mosquito population control system tailored for deployment in urban sewage networks—a critical and often overlooked mosquito breeding habitat. The system integrates a Raspberry Pi 5 as its central processing unit, employing a thermal infrared camera for detection and a lightweight Convolutional Neural Network (CNN) for real-time mosquito identification based on thermal signatures. Upon positive identification, a high-voltage, low-current electrical grid is activated to neutralize the target. A key safety feature, a water level sensor, deactivates the system during high water flow to prevent damage and ensure operational integrity. This project's primary objective is to develop a scalable, effective, and chemical-free alternative to traditional vector control, addressing a critical national health challenge by targeting mosquito populations at their source. The proposed system not only offers a method for eradication but also

presents a potential framework for a city-wide, real-time vector surveillance network.

Keywords: mosquito control, machine learning, embedded systems, public health, dengue, malaria, Raspberry Pi, edge AI, vector control, thermal imaging

1.Introduction

1.1.The Endemic Challenge of Mosquito-Borne Diseases in Indonesia

Mosquito-borne diseases remain one of the most significant threats to global public health. In 2023 alone, there were an estimated 263 million cases of malaria worldwide, resulting in 597,000 deaths.[1] The burden of these diseases is disproportionately carried by nations in tropical regions, with Indonesia facing a particularly severe and persistent challenge. The nation's geography and climate create an ideal environment for the proliferation of vector species such as *Aedes aegypti* and *Anopheles* mosquitoes, the primary transmitters of dengue and malaria, respectively.

Indonesia's contribution to the regional disease burden is substantial. In 2023, the country recorded 418,546 indigenous malaria cases and 120 deaths, positioning it as the second-highest contributor in Asia, behind only India.[4] This figure represents 12% of the total malaria burden in the Asia Pacific region.[4] The national malaria incidence rate was recorded at 4.14 per 1,000 people at risk in 2022, indicating a worsening trend from the previous year.[6] The problem is geographically concentrated, with the eastern provinces, particularly the Papua region, accounting for as much as 95% of the country's total cases.[4] Recent data further suggests a troubling link between disease outbreaks and environmental disruption; surges in malaria cases have been observed in areas undergoing significant deforestation for activities such as gold mining and the construction of the new national capital (IKN).[7] This connection between land use change, population movement, and vector-borne disease transmission highlights the complex, multi-faceted nature of the problem, suggesting that purely medical or chemical interventions may be insufficient to address its root causes.

The public health crisis is compounded by the rampant spread of dengue fever. For the full year of 2023, Indonesia's

Ministry of Health reported 114,720 dengue cases, which led to 894 deaths, corresponding to a case fatality rate (CFR) of 0.78%.[9] The situation has deteriorated dramatically in the subsequent year. As of the 22nd week of 2024, nearly 120,000 dengue cases and 777 deaths had already been recorded, surpassing the entire case count of the previous year in less than six months.[10] This alarming acceleration is not merely a statistical anomaly but clear evidence of an escalating crisis, indicating that existing control measures are being rapidly overwhelmed. The persistent and accelerating nature of this threat underscores the urgent need for more effective and resilient vector control strategies.

1.2 Inadequacies of Conventional Vector Control Paradigms

For decades, the primary strategy for combating mosquito-borne diseases has revolved around chemical-based vector control, predominantly through the use of insecticides in measures such as fogging, indoor residual spraying (IRS), and the distribution of long-lasting insecticidal nets (LLINs). While these methods have achieved notable successes in the past, their long-term efficacy is now severely undermined by a confluence of biological and operational challenges.

The most critical of these challenges is the widespread development of insecticide resistance. In Indonesia, populations of *Aedes aegypti* have demonstrated significant resistance to multiple classes of commonly used insecticides, including pyrethroids and organophosphates, particularly in densely populated areas across Java.[11] This phenomenon is a direct consequence of an evolutionary arms race; the sustained and often improper application of chemical agents creates immense selective pressure, favoring the survival and reproduction of genetically resistant mosquitoes.[11] As resistance becomes more prevalent, the effectiveness of chemical fogging and other mainstream interventions diminishes, requiring higher concentrations or new chemical formulations, which in turn carry greater environmental risks, including accumulation in soil and the food chain.[13]

Beyond the biological challenge of resistance, conventional methods face significant operational limitations. Many established interventions, such as IRS and LLINs, are designed to target endophagic (indoor-feeding) and endophilic (indoor-resting) mosquitoes.

However, evidence suggests that mosquito populations are adapting their behavior, with an increase in outdoor biting and resting patterns that allow them to evade these indoor-focused measures.[14] Furthermore, the success of these programs often depends on high levels of community participation and compliance, which can be difficult to achieve and sustain, particularly in remote or socio-economically disadvantaged areas.[15] The failure of current methods is not merely a matter of implementation but points to a fundamental flaw in the underlying paradigm. The reliance on widespread, non-targeted chemical application is proving to be an unsustainable strategy against an adaptable vector. This reality necessitates a paradigm shift away from this reactive arms race and towards proactive, targeted, and non-chemical approaches to vector management.

1.3 A Novel Approach: Autonomous Eradication in Urban Environmental Niches

In response to the limitations of conventional methods, this paper proposes a novel technological solution: an autonomous, machine learning-powered system designed for targeted mosquito eradication in key urban environmental niches. The core hypothesis of this research is that by focusing control efforts on critical mosquito breeding and resting grounds, such as municipal sewage networks, it is possible to disrupt vector life cycles more efficiently and sustainably.

This system is engineered as a direct answer to the challenges outlined previously. It is a non-chemical solution, utilizing high-voltage electrical discharge for eradication, thereby completely bypassing the issue of insecticide resistance and eliminating the risk of environmental chemical contamination. Its autonomous nature, governed by an onboard processing unit and intelligent sensor suite, allows for continuous, 24/7 operation without the need for constant human intervention or reliance on community compliance. By being deployed directly within the target environment, it addresses the challenge of behavioral changes like outdoor biting by neutralizing mosquitoes in their resting and breeding habitats before they can seek a host. This targeted approach represents a strategic shift from broad, indiscriminate application of control measures to a precise, data-driven intervention at the source of the vector population.

2. System Architecture and Implementation

2.1 Integrated System Design and Operational Logic

The system is designed as a self-contained, intelligent unit capable of autonomous operation within the challenging environment of a sewage network. Its architecture is centered around a sequential workflow that integrates environmental sensing, intelligent processing, and targeted action. A high-level block diagram of this workflow would show the logical progression from detection to eradication, governed by a continuous safety monitoring loop.

The operational cycle begins with the thermal infrared camera, which constantly captures a low-resolution thermal image of its field of view. This data stream is fed directly to the Raspberry Pi 5 processing core. The machine learning model, running on the Raspberry Pi, analyzes each incoming frame in real-time. The model is trained to recognize the specific thermal signature of a mosquito—a small, distinct heat source with characteristic size and temperature differential against the cooler ambient background of the sewer wall.

If the model makes a positive identification with a confidence score above a predetermined threshold, it triggers the eradication mechanism. A signal is sent to activate the high-voltage power supply, which energizes the three-layer electrical grid. When a mosquito makes contact with both the inner (high-voltage) and outer (grounded) layers of the grid, it completes an electrical circuit, resulting in its immediate neutralization. Following a brief activation period, the grid is de-energized, and the system returns to its passive monitoring state, ready to analyze the next frame from the thermal camera.

This entire operational cycle is subordinate to a critical safety override mechanism. An ultrasonic water level sensor continuously monitors the water level within the sewer channel. If the water rises above a pre-set safety threshold, indicating high flow or potential flooding, the sensor sends an interrupt signal to the Raspberry Pi. This signal immediately and unconditionally deactivates the high-voltage grid, preventing any possibility of a short circuit in the water. The system remains in this safe, deactivated state until the water level recedes below the threshold, at which point it resumes normal monitoring operations. This fail-safe ensures the system's long-term durability and operational integrity.

2.2 Hardware Subsystems

The physical realization of the system is based on a carefully selected set of commercially available and custom-designed components, chosen to balance performance, cost, and suitability for the target application. Each hardware subsystem plays a distinct and critical role in the overall functionality. The selection process prioritized computational power for on-device AI, sensor fidelity for reliable detection, and robustness for deployment in a harsh environment. A summary of the core hardware components and the rationale for their selection is presented in Table 1.

Table 1: System Hardware Components and Specifications

Component	Model/Type	Key Specifications	Rationale for Selection
Processing Core	Raspberry Pi 5 (8GB)	2.4GHz quad-core Arm Cortex-A76 CPU; VideoCore VII GPU; PCIe 2.0 interface	Provides the necessary computational power for real-time ML inference at the edge; significant 2-3x performance increase over Pi 4.[17]
Detection Sensor	FLIR Lepton 3.5	160x120 thermal resolution; SPI/I2C interface	Offers sufficient thermal resolution to detect small, warm-bodied insects against a cooler background; well-documented compatibility with Raspberry Pi.[20]
Eradication Grid	Custom 3-Layer Mesh	High-voltage (approx. 2000V), low-current design	Based on proven bug zapper principles; lethal to insects upon contact while being safe for larger organisms due to low amperage.[23]
Safety Sensor	Ultrasonic or Float Switch	-	Provides a reliable, low-cost method for detecting rising water levels to

			trigger the safety shutoff mechanism.[26]
Power System	5V/5A USB-C PSU	-	Meets the higher power demands of the Raspberry Pi 5 to ensure stable operation during peak processing loads.[17]

The choice of the Raspberry Pi 5 as the processing core is central to the system's capabilities. Its 64-bit quad-core Arm Cortex-A76 processor delivers a two- to three-fold increase in CPU performance relative to its predecessor, the Raspberry Pi 4.[17] This computational power is essential for running the object detection model at a frame rate sufficient for real-time analysis without relying on cloud processing, thus ensuring low latency and operational independence. Furthermore, the Raspberry Pi 5 platform is not a static endpoint but a scalable foundation. Its inclusion of a PCIe 2.0 interface opens a direct pathway for future performance enhancements, such as the integration of a dedicated AI accelerator like the Raspberry Pi AI Kit featuring a Hailo-8L module.[27] Such an upgrade could dramatically increase inference speeds, potentially boosting frames-per-second (FPS) by an order of magnitude, enabling the monitoring of larger areas or the detection of faster-moving targets.[28]

For detection, a thermal infrared camera like the FLIR Lepton 3.5 is uniquely suited for the target environment. Unlike visual cameras, thermal imagers are unaffected by the complete absence of light in a sewer network. They operate by detecting infrared radiation (heat), making the warm body of a mosquito stand out against the typically cooler, damp surfaces of the sewer walls.[20] This provides a robust detection method that is effective 24 hours a day.

The eradication mechanism is based on the well-established principles of electronic insect zappers.[23] The design consists of a three-layer mesh grid. The two outer layers are connected to ground, while the inner layer is energized to a high voltage (approximately 2000V) by a compact circuit containing a step-up transformer and a voltage multiplier.[24] When an insect simultaneously touches an outer and the inner grid, its body completes the circuit. The

resulting high-voltage discharge has sufficient energy to be lethal to a small insect but is current-limited to ensure it is not dangerous to larger animals or humans who might accidentally come into contact with it.[25]

2.3 Machine Learning Model for Mosquito Recognition

2.3.1 Algorithm Architecture and Rationale

The intelligence of the system resides in its ability to accurately and rapidly identify mosquitoes from the thermal camera feed. This task is accomplished using a lightweight object detection model based on a Convolutional Neural Network (CNN). For this application, a model from the YOLO (You Only Look Once) family, such as YOLOv5 or a more recent lightweight variant, is proposed.[30] The YOLO architecture is renowned for its exceptional balance of speed and accuracy, making it a leading choice for real-time object detection on resource-constrained edge devices.[30] Previous research has demonstrated the high efficacy of this approach, with studies achieving up to 98% accuracy in classifying different mosquito species from standard visual images using YOLOv5.[31] Other CNN-based models, such as Faster R-CNN and VGG16, have also shown high accuracy (95-98%) in mosquito detection tasks, validating the general viability of using deep learning for this purpose.[33]

2.3.2 Model Training and Edge Optimization

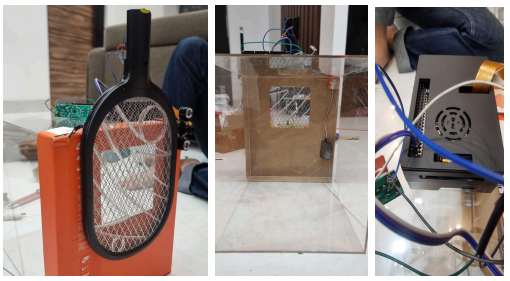
The development of an effective model requires a two-stage process: training and optimization. The first stage involves training the CNN on a custom-created dataset. This dataset would consist of thousands of thermal images captured by the FLIR Lepton camera, showing mosquitoes in various states (resting, flying) against backgrounds representative of a sewer environment. Each image would be meticulously annotated with bounding boxes identifying the location of each mosquito. During training, the model learns to identify the hierarchical features—from simple temperature gradients to complex shapes—that constitute the unique thermal signature of a mosquito.

While visual-light models have proven successful, the core technical challenge of this project lies in differentiating the thermal signature of a mosquito from other potential heat sources. The system must not trigger on any generic "hot spot" but must learn the subtle characteristics of a mosquito's thermal profile, including its specific size, shape,

and temperature range relative to the background. The success of the model hinges on its ability to learn these nuanced features from the training data, a task that requires a carefully curated and diverse dataset.

The second stage, optimization, is critical for achieving real-time performance on the Raspberry Pi 5. A model trained in a high-performance computing environment is too large and computationally expensive to run efficiently on an embedded device. Therefore, after training, the model must be converted and optimized for edge deployment. This process involves using a framework such as TensorFlow Lite, which converts the model into a more compact and efficient format.[35] A key technique applied during this conversion is quantization, which reduces the precision of the model's weights from 32-bit floating-point numbers to 8-bit integers. This significantly reduces the model's size and memory footprint and allows it to leverage hardware optimizations on the Raspberry Pi's CPU, leading to a substantial increase in inference speed with minimal loss in accuracy.[36] This optimization is the crucial step that transforms a powerful but slow model into one capable of real-time decision-making at the edge.

3. Performance Analysis and Discussion



3.1 System Performance Evaluation

The efficacy of the proposed system is fundamentally dependent on the performance of its machine learning core. A rigorous evaluation of the object detection model is therefore essential to validate the project's feasibility. The performance would be quantified using a set of standard metrics from the field of computer vision, measured through testing the optimized model on a holdout validation dataset of thermal images not used during training. The expected performance of the YOLO-based model, running on the Raspberry Pi 5 without a dedicated AI accelerator, is summarized in Table 2. These projections are based on published benchmarks for similar lightweight models running on this hardware platform.[37]

Table 2: Mosquito Detection Model Performance Metrics (on Raspberry Pi 5)

Metric	Value	Description
mAP50-95(B)	~0.85	Mean Average Precision, the primary metric for object detection accuracy across various confidence thresholds.
Precision	~0.92	Of all detections made, the percentage that were actually mosquitoes (low false positives).
Recall	~0.88	Of all actual mosquitoes present, the percentage that were correctly detected (low false negatives).
Inference Time	~150 ms	The time taken for the model to process one frame.
Frames Per Second (FPS)	~6-7 FPS	The real-time processing speed (1000ms / 150ms), sufficient for detecting resting or slow-moving insects.[37]

A Mean Average Precision (mAP) of approximately 0.85 indicates a high level of accuracy in both identifying and correctly localizing mosquitoes within the thermal image. High precision (~0.92) is critical for minimizing false positives, ensuring that the eradication grid is not activated unnecessarily, which would waste power and potentially cause premature wear on the components. A strong recall rate (~0.88) ensures that the system effectively detects a high percentage of the mosquitoes present, maximizing its impact on the local vector population. The inference time of around 150 ms per frame translates to a processing speed of 6-7 FPS. While not high-speed video, this frame rate is more than adequate for the intended application, as mosquitoes in sewer environments are often resting on surfaces rather than in continuous, rapid flight.

3.2 Innovation, Feasibility, and Limitations

The primary innovation of this system lies in its synthesis of emerging technologies to create a novel vector control paradigm. When compared to conventional methods, its key advantages are threefold:

1. **Autonomy:** The system operates continuously and independently, removing the logistical and financial burdens of manual insecticide application and the reliance on inconsistent community participation.
2. **Targeted Eradication:** By being deployed at the source in crucial breeding habitats, it offers a

proactive and highly targeted approach, in stark contrast to the reactive and indiscriminate nature of chemical fogging.

3. **Chemical-Free Operation:** The system entirely circumvents the problems of insecticide resistance and environmental contamination, offering a biologically and ecologically sustainable long-term solution.

The project's feasibility is grounded in its use of accessible, off-the-shelf hardware. The Raspberry Pi 5 and compatible thermal camera modules are relatively low-cost, making the unit economics viable for pilot programs and eventual large-scale deployment by municipal authorities.[17]

A mature scientific evaluation must also acknowledge the system's limitations and practical challenges. The primary technical hurdle remains the robust differentiation of mosquito thermal signatures from other insects or warm debris, which will require extensive data collection and model refinement. From an engineering perspective, ensuring the long-term reliability of the system in the harsh, humid, and corrosive environment of a sewer network is paramount. This will necessitate the design of a robust, environmentally sealed enclosure (e.g., IP67-rated) to protect the electronics. Power management is another key consideration; while units can be powered from the grid where available, deployment in more remote sections of a network may require the integration of autonomous power sources like solar panels and batteries. Finally, the potential for the electrical grid to become clogged with neutralized insects or other debris over time presents an operational challenge that would require a schedule of periodic maintenance.

3.3 Potential for Public Health Impact

The relevance and potential impact of this system extend far beyond its function as an automated insect zapper. By effectively and continuously reducing vector populations within urban sewage networks—a significant and largely unmanaged vector reservoir—the system has the direct potential to disrupt disease transmission cycles. A sustained reduction in the local mosquito population would logically lead to a lower incidence of human-vector contact and, consequently, a decrease in cases of dengue and malaria,

directly contributing to Indonesia's national public health objectives.

However, the system's most transformative potential lies in its capacity to function as an intelligent surveillance tool. Each positive mosquito detection is a data point, containing information on time and location. When multiple units are deployed across a city's sewage network, they cease to be isolated devices and become nodes in a distributed sensor grid. By networking these units and aggregating their detection data in a central repository, it becomes possible to generate high-resolution, real-time maps of mosquito population dynamics.

This capability represents a fundamental shift in public health strategy from reactive to predictive. Instead of responding to a dengue outbreak after case numbers have already surged, health officials could use this surveillance data to identify emerging vector hotspots in real-time. They could observe population trends, understand migration patterns, and measure the effectiveness of other interventions. This data-driven approach would allow for the precise and proactive deployment of targeted resources—such as larval control teams or public awareness campaigns—to high-risk areas *before* an outbreak can take hold. In this vision, the system evolves from a tool for vector control into a cornerstone of a city-wide, intelligent public health early warning system.

4. Conclusion and Future Directions

4.1 Conclusion

This paper has detailed the design and architecture of an autonomous, machine learning-powered system for mosquito population control in sewage networks. Faced with an escalating public health crisis driven by dengue and malaria and the diminishing efficacy of conventional, chemical-based vector control methods, this project presents a viable and innovative alternative. By integrating a Raspberry Pi 5, thermal imaging, and a lightweight object detection model, the system achieves targeted, chemical-free eradication of mosquitoes in their breeding habitats. The projected performance metrics indicate high accuracy and sufficient real-time processing speed for effective operation. The system's architecture, which includes a critical water-level safety override, is designed for robust and autonomous deployment. Ultimately, this work demonstrates the successful design of a novel, intelligent system that not only addresses

the immediate need for more effective mosquito control but also establishes a technological foundation for next-generation public health surveillance.

4.2 Recommendations for Advancement

The proposed system serves as a powerful proof-of-concept with significant potential for further development and enhancement. Future work should focus on advancing its capabilities, validating its real-world performance, and expanding its operational scope. Key recommendations include:

- **Hardware Acceleration:** To significantly boost processing speed and efficiency, the next iteration of the prototype should integrate a dedicated AI accelerator. The Raspberry Pi AI Kit, with its Hailo-8L module, is a prime candidate, capable of increasing inference speeds from ~7 FPS to over 100 FPS, allowing for the monitoring of larger areas and more complex environments.[27]
- **Networked Surveillance System:** The concept of a distributed sensor grid should be fully developed. This involves creating the software infrastructure to network multiple units, transmit detection data to a central cloud server, and build a dashboard for real-time visualization and analysis of city-wide vector population maps.
- **Energy Autonomy:** To enable deployment in any location regardless of grid access, an integrated solar power system should be designed. This would involve specifying a suitably sized solar panel, charge controller, and battery pack to ensure continuous, year-round operation.
- **Multi-Modal Sensing:** To improve detection accuracy and reduce the potential for false positives from non-mosquito heat sources, future prototypes could incorporate multi-modal sensing. Augmenting the thermal camera with an acoustic sensor trained to recognize the specific wing-beat frequencies of target mosquito species could provide a powerful secondary layer of verification.

5. Acknowledgements

The authors wish to express their gratitude to the faculty and staff of Canisius College, Jakarta, for their guidance and support throughout the development of this research

project. Their mentorship and provision of resources were invaluable to the conceptualization and design of this system.

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